

YIJIA WANG

Seattle, WA • yijia0725@gmail.com • 276-870-4745 • **U.S citizen** - No sponsorship required
[Personal Website](#) • [LinkedIn - Yijia Wang](#) • [GitHub - yijia0725](#) • [Project - Housing Analytics](#)

Data Scientist with an M.S. in Applied Statistics (UMich) and a strong foundation in translating complex data into actionable business strategies. Expertise in advanced SQL, Machine Learning, and data pipeline automation. Proven ability to optimize operational workflows and partner effectively with technical and non-technical stakeholders to drive data-informed decisions.

SKILLS

- **Statistical Methods:** Experimental Design(A/B Testing), Statistical Modeling, ANOVA/Chi-Square, Bayesian Statistics, ARIMA
- **Programming Languages:** Python(PyTorch, Tensorflow, Pandas, Numpy, Seaborn), SQL, NoSQLR
- **Machine Learning:** Tree-based Models (XGBoost, Random Forest), Clustering (K-Means), NLP
- **Statistical Modeling:** Linear & Logistic Regression, LDA
- **Data Processing:** Data mining, Feature Engineering, Imbalanced Data Handling (SMOTE), Data Structures, Data Warehousing
- **Tools:** Tableau (Data Visualization, Dashboard), Google Looker Studio, Excel, SAS, BigQuery (ETL), Streamlit, Git, Web Scraping

EXPERIENCE

Data Scientist *Open House Atlanta Realty & Investment* Nov 2024 – Feb 2026

- Implemented an **automated pipeline** that executes batched Propertyware API calls, generates aggregated maintenance request tables.
- Maintains data integrity via robust backup logs and cell-wise updates, reducing operation latency for 15 managers **from >1 day to 2 hrs.**
- Architected a performance bonus system for 50+ Make-Ready employees, aligning 7 department heads on statistical criteria.
- Designed new **data structures** and **integrated** dispersed operational metrics into the **centralized data warehouse.**
- **Synthesized** multi-source data into transparent, audit-ready evaluation sheets to **drive employee incentives.**
- Designed interactive Looker Studio **dashboards** via **BigQuery (SQL)** to streamline data analysis, tracking **key performance indicators** for operational and financial analysis (Occupancy, Rent Collection), **reducing time-to-insight by 40%.**
- Engineered a scalable **Streamlit (Python)** analytics web app for house repair procurement with **automated** Buildertrend web scraping and secure authentication, **replacing** fragmented **manual logs** to enable rapid retrieval of multi-year records.
- Developed an **NLP pipeline** using TF-IDF and LDA to analyze 500+ natural language renovation purchase orders from Renowalk; Identified high-volume item **bottlenecks** to project a 15% reduction in inventory storage overhead.

Student Researcher - Liver Disease Analytics *UMich mentored by Dr. Jessica Mellinger* Aug 2023 – Dec 2023

- Processed ~200 patient demographic & medical records.
- Conducted **hypothesis testing** and logistic **regression** for disease outcome **prediction in R.**
- Authored a reproducible report to optimize future clinical data collection strategies.

Research Assistant - High Throughput Phenotyping *Virginia Tech mentored by Dr. Song Li* Nov 2021 – Mar 2022

- Executed data acquisition for computer vision-based crop phenotyping to automate edamame quality assessment.
- Annotated precise bounding boxes using VGG Image Annotator on hundreds of samples to establish ground-truth labels for model training.

Graduate Student Instructor (Statistics) *UMich* Aug 2022 – May 2024

- Taught 300+ students across 5 semesters on **statistical theory** using R and Python.
- Led weekly lab discussions, developed instructional slides, designed homework problems, held office hours, and graded assignments.

Teaching Assistant *The Coding School* Jul 2023 – Aug 2023

- Answered questions and facilitated small group discussions for high school students in a college-level Intro to R course.

Math Tutor *Mountain Empire Community College* Mar 2019 – Apr 2020

- Provided 1-on-1 tutoring for entry-level math classes (Algebra, Calculus).

EDUCATION

University of Michigan, Ann Arbor, MI May 2024

MS in Applied Statistics GPA: 3.65/4.0 - [Transcript](#)

Virginia Tech, Blacksburg, VA May 2022

BS in Statistics, Minor in Mathematics GPA: 3.96/4.0 - [Transcript](#)

- **Featured Courses:** Causal Inference, Induction, Theoretical Statistics, Experimental Design, Regression Analysis, Bayesian Modeling, Time Series Analysis, Database App Design

- **Honors:** Academic Dean's List; Whitfield Cobb Award for Academic Achievement